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Long-Period Variables in the Large Magellanic Cloud

Michele Trabucchi,[1] Peter R. Wood,[2] Josefina Montalbán,[1] Paola Marigo,[1] Giada Pastorelli,[1] and Léo Girardi[3]
[1]*University of Padova, Department of Physics and Astronomy "Galileo Galilei", Padova, Italy;* michele.trabucchi@unipd.it

[2]*Research School of Astronomy and Astrophysics, Australian National University, Canberra, ACT, Australia*

[3]*Astronomical Observatory of Padova - INAF, Padova, Italy*

Abstract. We present the first results from a new set of linear, radial, non-adiabatic pulsation models of long-period variables (LPVs), widely covering the space of stellar parameters corresponding to evolution along the Asymptotic Giant Branch (AGB). This coverage includes several values of metallicity and carbon-enhanced compositions consistent with the occurrence of repeated third dredge-up events. Updated atomic and molecular opacity data were included consistently with chemical composition in order to produce realistic pulsation properties of both O-rich and C-rich variables. We combined the pulsation models with a stellar population synthesis tool in order to study the population of LPVs in the Large Magellanic Cloud (LMC). We found the models to be reasonably good at reproducing the observed period–luminosity (PL) sequences, as well as the distribution of observed amplitudes. We provide a new interpretation of the PL sequences of LPVs that can bring into alignment previously discordant interpretations.

1. Introduction

Luminous red giant stars exhibit variability with periods between a few days to more than a thousand days. These stars are either approaching the tip of the Red Giant Branch (RGB) or they are evolving along the Asymptotic Giant Branch (AGB; Ita et al. 2002; Kiss & Bedding 2003), and are known as long-period variables (LPVs). They are characterised by following rather defined relations in the period–luminosity (PL) diagram, as first shown by Wood et al. (1999) and Wood (2000), who identified five PL relations (or sequences) in the LMC and labelled them with letters A, B, C, D and E. Later studies (Ita et al. 2004; Soszyński et al. 2004) provided a more detailed picture, showing that: (1) sequence B is actually formed by two parallel sequences, labelled B and C′, (2) an additional sequence (A′) exists on the short-period side of sequence A, and (3) sequences A and B, and possibly A′, consist of three or more closely spaced ridges.

The existence of such closely spaced ridges is likely due to non-radial oscillation. Here, we only discuss the implications from radial pulsation models, while results from non-radial oscillation models of luminous red giants are discussed by Montalbán et al. (in preparation).

Sequences A′, A, B, C′ and C are due to stellar pulsation, while sequence D is associated with the phenomenon of long secondary periods (LSPs), which is still not understood. Sequence E is due to binary stars and not further discussed here. Fig. 1 provides an example of the appearance of the PL sequences of LPVs in the LMC based on data from the OGLE III Collection of Variable Stars (OGLE-III CVS; Soszyński et al. 2009).

The existence of several distinct PL sequences is attributed the fact that LPVs pulsate in a few different modes of low radial order. It is usually assumed that the observed sequences correspond to distinct and adjacent radial orders of pulsation. A number of attempts have been made to identify the modes responsible for each sequence, as clearly summarised by Wood (2015). Nonetheless, there is still some disagreement in terms of the mode assignment. Indeed, two main interpretations, incompatible with each other, are present in literature.

One of the two interpretations (supported, e.g., by Wood 2000; Soszyński et al. 2013; Takayama et al. 2013; Wood 2015) assumes that sequences C, containing the Mira variables, is due to pulsation in the radial fundamental mode, while sequences C′, B, A, and A′ correspond to the radial first, second, third and fourth overtone modes, respectively.

On the other hand, according to the interpretation supported, e.g., by Soszyński et al. (2007), Dziembowski & Soszyński (2010), and Mosser et al. (2013), sequences B and A correspond to the first and second radial overtone modes, so that A′ would be due to the third overtone mode and sequence C′ would be associated with the fundamental mode. This way, however, there would be no explanation for sequence C in terms of radial pulsation, although Mira variables (generally accepted to be fundamental mode pulsators) are found in this sequence. There is thus a one-sequence offset between the mode assignments corresponding to the two interpretations discussed above.

Here we present a new interpretation based on the combination of stellar population synthesis with a large set of new pulsation models of LPVs. The results, supported by a semi-empirical analysis, suggest that the extant disagreement in the mode assignments can be solved by relaxing the requirement that adjacent observed sequences correspond to distinct and adjacent radial modes.

2. Synthetic Population of LPVs

The details of the theoretical modelling involved in the simulation of a population of LPVs can be found in Trabucchi et al. (2017). In summary, we used a synthetic stellar population representative of the LMC computed with the `TRILEGAL` code (Girardi et al. 2005). Then, we assigned periods and growth rates for five radial modes to each simulated RGB and AGB star, using the pulsation models described by Trabucchi et al. (in preparation), computed with the code described by Wood & Olivier (2014). Pulsation models account for enhancement in the envelope abundance of carbon due to the third dredge-up, and include detailed atomic and molecular opacities computed with the `AESOPUS` code (Marigo & Aringer 2009), in order to provide a realistic description of pulsating carbon stars.

We compared the synthetic population of LPVs so obtained with data from the OGLE-III CVS, complemented with near-infrared JHK_{s} photometry from the Two Micron All Sky Survey (2MASS; Skrutskie et al. 2006). The OGLE-III catalogue includes three observed periods, and corresponding I-band amplitudes, for each star. The most

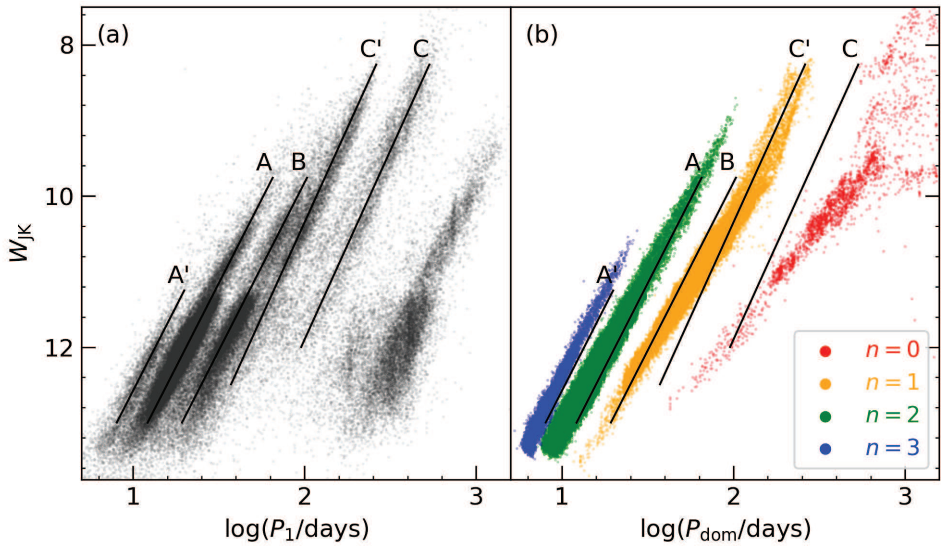

Figure 1. Observed primary periods (a) compared to dominant periods obtained from radial, linear, non-adiabatic pulsation models (b), where different colours represent different radial orders of pulsation: the fundamental mode (red), the first overtone mode (orange), the second overtone mode (green) and the third overtone mode (blue). Solid lines provide a visual aid to identify the approximate position and slope of the observed PL sequences.

significant observed period for each star is called the primary period, and it usually has a larger amplitude than the other two, the secondary and tertiary periods. The theoretical counterpart of the observed primary periods is represented by the pulsation mode, for each simulated star, characterised by the largest growth rate.

3. Results

We compared observed primary periods with theoretical dominant periods in the $\log(P)$–W_{JK} plane (Fig. 1), where the Wesenheit index W_{JK} is a reddening-free measure of luminosity defined by

$$W_{JK} = K_{\rm s} - 0.686(J - K_{\rm s})\,. \tag{1}$$

We find the models to be in reasonably good agreement with the observations. The period distributions of dominant third and second overtone modes from the models are able to reproduce both the slope and position of sequences A′ and A reasonably well. The distribution of dominant first overtone periods in the models covers the region occupied by both sequences B and C′ in the PL diagram, but does not reproduce the splitting into two separate sequences.

On the other hand, theoretical dominant fundamental mode periods are in poor agreement with observed primary periods in sequence C. To understand the reasons for this disagreement, we examined the distributions of observed amplitudes.

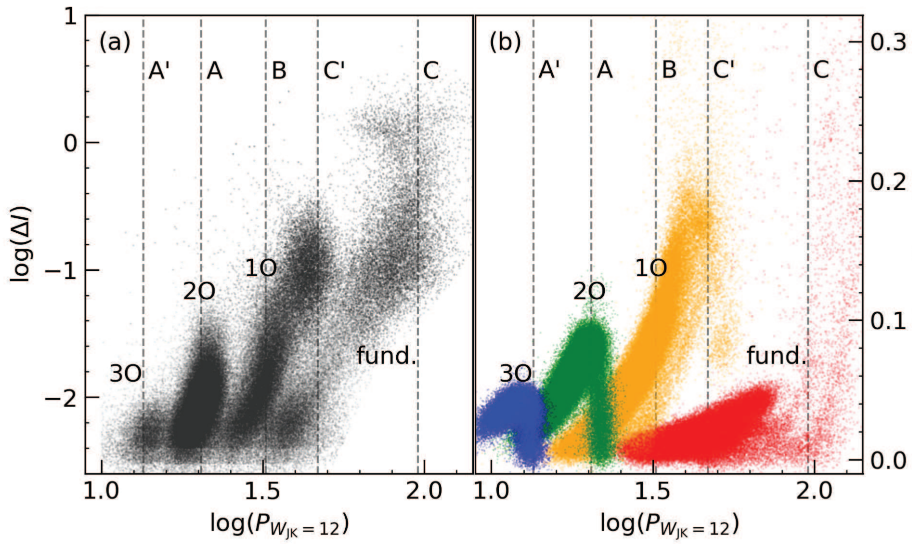

Figure 2. (a) Observed I-band amplitudes for all three periods (primary, secondary and tertiary) of LPVs in the OGLE-III CVS, as a function of the projected period $\log(P_{W_{JK}=12})$. The observed PAL relations are labelled with the corresponding mode according to the interpretation provided here (see text). (b) Theoretical growth rates for all excited modes up to the third overtone, labelled and colour coded according to their radial order as in Fig. 1. Dashed lines in the background approximately identify the centre of the observed PL sequences.

This is shown in Fig. 2, where the quantity on the horizontal axis is the observed $\log(P)$ projected along a line with the same slope of sequence A:

$$\log(P_{W_{JK}=12}) = \log(P) + (W_{JK} - 12)/4.444\,. \tag{2}$$

By using the "projected periods" we are able to examine the behaviour of amplitudes across the PL sequences (as explained in more detail by Trabucchi et al. 2017; see also Wood 2015). We found data from observations to form four distinct period–amplitude–luminosity (PAL) relations, that we interpreted as corresponding to four distinct radial orders of pulsation, from the fundamental mode to the third overtone mode.

To support this interpretation, we used theoretical growth rates as a proxy for variability amplitudes, and found that they exhibit distributions very similar to those of the observed amplitudes, at least for the overtone modes. This correspondence supports the idea that sequences A′ and A are due to pulsation in the radial third and second overtone modes, respectively, while both sequences B and C′ correspond to the radial first overtone mode, as suggested by the comparison of models and observations in the PL diagram in Fig. 1.

Again, we found the theoretical fundamental mode to be in poor agreement with the observations. However, the disagreement is largely due to the growth rates, while fundamental mode periods predicted by models are roughly in agreement with observations. In other words, the main issue in modelling the fundamental mode is connected to the description of the driving. This can be seen by comparing theoretical and observed fundamental mode periods regardless of their degree of excitation.

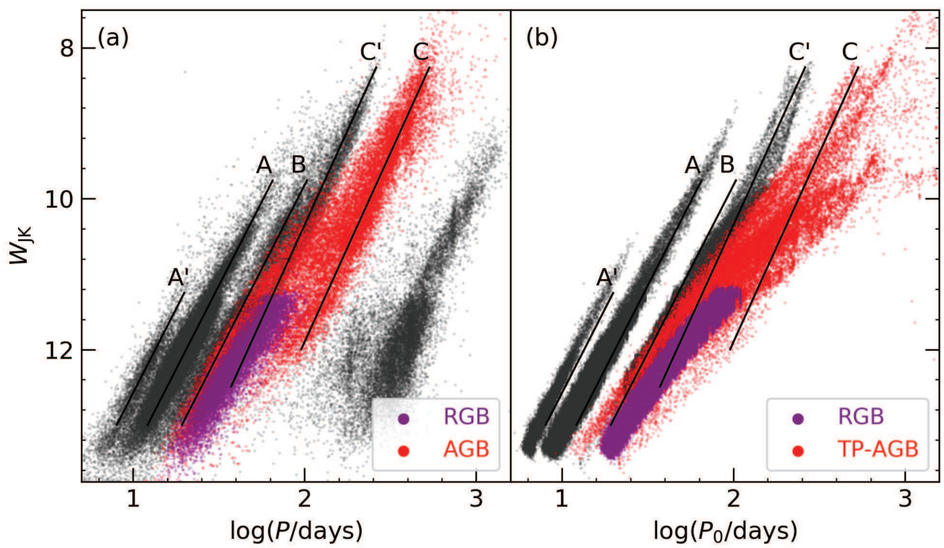

Figure 3. (a) Observed PL diagram of stars from the OGLE-III CVS. Grey points are observed primary periods. Observed fundamental mode periods are shown as red (AGB) and purple (AGB) points. These include any one of the three periods (primary, secondary and tertiary) of OGLE-III stars, provided that they belong to the PAL relation labelled "fund." in Fig. 2. Panel (b): theoretical PL diagram. Grey points are dominant overtone mode periods. The fundamental mode, regardless of growth rate, is shown as red (TP-AGB) and purple (RGB) points.

This means that all theoretical fundamental mode periods are taken into account, regardless of the growth rates. As for the observed periods, we ignore their amplitudes, and include any one of the three periods of OGLE-III stars (primary, secondary or tertiary) provided that they belong to the PAL relation we identified as fundamental mode pulsation (see Fig. 2). The comparison (Fig. 3) displays a better agreement, with both the observed and theoretical distributions extending from sequence B to sequence C.

These results suggest that the PL sequences of LPVs in the LMC can be explained in terms of pulsation in exactly four radial orders, from the fundamental to the third overtone mode. As discussed by Trabucchi et al. (2017), this picture is entirely supported by the analysis of the distribution of multiple periods in individual stars. According to this interpretation, both sequences B and C′ are due to the same pulsation mode, i.e., the first overtone. Models are unable to reproduce the observed splitting of the two sequences, as it is due to the presence of long secondary periods.

In fact, stars that have a LSP on sequence D have often another period between sequences B and C′. If the LSP in those stars is classified as primary period, then the corresponding period between sequences B and C′ will be classified as secondary or tertiary (this happens for about 40% of stars with a period on sequence D). Usually, only primary modes are shown in observed PL diagrams. As a consequence, all periods between sequences B and C′ corresponding to a LSP classified as primary do not appear in PL diagrams, giving rise to the apparent gap between those two sequences. In fact, if the classification of observed periods into primary, secondary and tertiary is performed after removing all LSPs from the data set, the splitting disappears, as shown in Fig. 4.

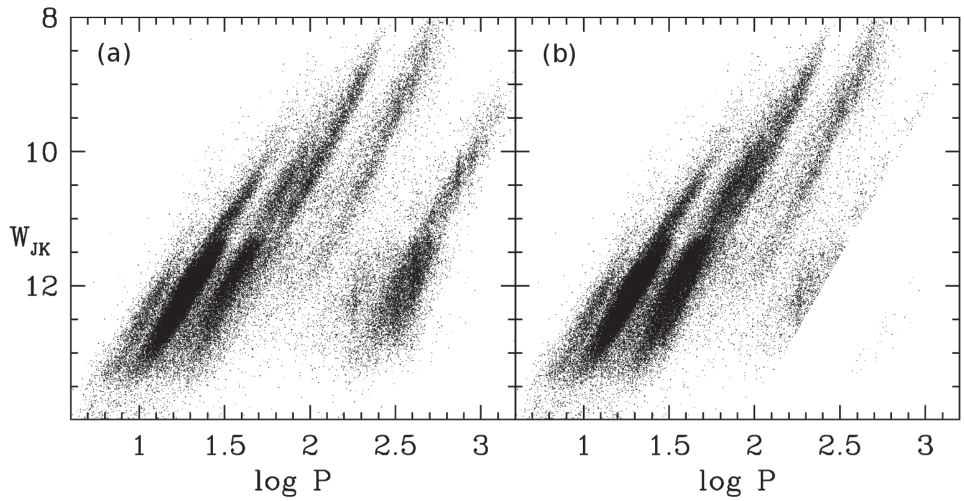

Figure 4. (a) PL diagram of observed primary periods. (b) PL diagram of observed periods corresponding to the largest amplitude, after excluding LSPs on sequence D.

4. Conclusions

We used linear, radial, non-adiabatic pulsation models to simulate the pulsation properties of long-period variables in a synthetic stellar population representative of the Large Magellanic Cloud. We compared the results with OGLE-III observations and found reasonably good agreement for the predicted properties of overtone modes, both in terms of periods and growth rates. In particular, the latter provide a good proxy for variability amplitudes.

We found that the observed PL sequences of LPVs in the LMC can be explained by pulsation in exactly four radial orders, with sequences A′ and A corresponding to the radial third and second overtone modes, sequences B and C′ being both due to the first overtone mode, and sequence C corresponding to the fundamental radial mode.

This interpretation is able to bring into alignment the mode assignment of Soszyński et al. (2007), Dziembowski & Soszyński (2010), and Mosser et al. (2013), and the one supported by Wood (2000), Soszyński et al. (2013), Takayama et al. (2013), and Wood (2015).

The observed gap between sequences B and C′ is not reproduced by models. It must be attributed to the occurrence of long-secondary period variability (which nature is unknown) in stars having a dominant first overtone mode with a period between sequences B and C′. A significant fraction of such stars have LSPs with larger amplitude than the first overtone mode, so that the former is classified as primary. This leads to the apparent splitting of the distribution of first overtone periods into the two sequences B and C′. Since the pulsation models do not include a description of LSPs, they are also unable to produce a splitting in the distribution of dominant first overtone mode periods.

The above interpretation is supported by empirical evidence in terms of the observed amplitudes, as well as the distribution of multiple periods in individual stars in the PL diagram (Trabucchi et al. 2017; see also Wood 2015). According to this mode assignment, there is no one-to-one correspondence between observed sequences and radial orders of pulsation. On the other hand, observed LPVs form four distinct PAL relations, each one corresponding to a different radial order of pulsation.

While models provide valid predictions in terms of pulsation in the overtone modes, they are not able to properly describe the driving of the fundamental mode. As a consequence, the region in the PL diagram where the fundamental mode is predicted to be dominant (expected to correspond to observed primary periods on sequence C) is in poor agreement with the observations. However, such a discrepancy is mainly due to the unreliable values of the fundamental mode growth rates, while periods are roughly compatible with observations. To verify this, we examined the distribution in the PL diagram of theoretical fundamental mode periods regardless of their growth rates. We compared them to observed periods corresponding to the PAL relation identified as fundamental mode pulsation, including primary periods as well as secondary and tertiary. The comparison shows moderately good agreement although models tend to overestimate the period of the fundamental mode, especially towards high luminosity.

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